INCH-POUND

A-A-60051

6 December 2022

COMMERCIAL ITEM DESCRIPTION

PLUGS, PIPE AND PORT, CADMIUM AND HEXAVALENT-CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers plugs which shall not contain cadmium or hexavalent-chromium in their finishes.

2 CLASSIFICATION

2.1 Types.

Type I	-	SAE AS4863 - Plug, Pipe, External Thread
Type II	-	SAE AS5169 - Fitting, Port Plug and Bleeder

3 SALIENT CHARACTERISTICS.

3.1 Materials. Plugs shall be manufactured as specified in the Society of Automotive Engineers (SAE) documents listed in 2.1 by type, except the plating/finishes shall not contain cadmium or hexavalent-chromium as specified herein (see 3.3).

3.2 Dimensions. Dimensions for all designated type of plugs apply after plating/finishing. Refer to each individual SAE specification listed in 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent-chromium free Part or Identifying Number (PIN) codes and finish/plating options. PIN finish/plating codes for all designated types of plugs:

QW	-	Steel shall be Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K20.
P5	-	CRES shall be cleaned, descaled and passivated in accordance with SAE AMS2700 Method 1, Type 6, 7, or 8, or Method 2, or ASTM A967/A967M. Post-cleaning treatment is prohibited.
J	-	Aluminum / aluminum alloy shall be chemically treated in accordance with MIL-DTL-5541, Type II, Class 1A or shall be anodized in accordance with MIL-PRF-8625, Type IC, II or IIB, Class 1 or 2. Sealers (clear or any color) containing hexavalent chromium shall be prohibited.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:33Z
Check the source to verify that this is the current version before use.

3.4 Interchangeability. When SAE specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent-chromium with the cadmium and hexavalent-chromium free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections.

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this CID and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. The following finish/plating PIN's are cadmium and hexavalent-chromium free for the designated plugs.

TABLE I. Type I (-1), plug, pipe, external thread.

New A-A-60051 Part Number	Reference Dash Number (SAE AS4863)
AA60051-1-( ) 00 (X)	AS4863 ( ) 00 (FC)
AA60051-1-( ) 02 (X)	AS4863 ( ) 02 (FC)
AA60051-1-( ) 03 (X)	AS4863 ( ) 03 (FC)
AA60051-1-( ) 04 (X)	AS4863 ( ) 04 (FC)
AA60051-1-( ) 06 (X)	AS4863 ( ) 06 (FC)
AA60051-1-( ) 08 (X)	AS4863 ( ) 08 (FC)
AA60051-1-( ) 10 (X)	AS4863 ( ) 10 (FC)
AA60051-1-( ) 12 (X)	AS4863 ( ) 12 (FC)
AA60051-1-( ) 16 (X)	AS4863 ( ) 16 (FC)
AA60051-1-( ) 20 (X)	AS4863 ( ) 20 (FC)

(FC) = Finish codes from SAE AS4863, differ from this CID's finish codes unless otherwise indicated.

Table notes: (continued on next page)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:33Z
Check the source to verify that this is the current version before use.

PIN construction

(Copper alloy is not included in this CID, see SAE AS4863)

AA60051 -1- ( ) 04 (X) A

- The "A" indicates for use in oxygen systems (see SAE AS4863).
- Finish code (see 3.3).
- Size code, fitting end sizes in 0.125-inch increments (see SAE AS4863).
- Material codes (from SAE AS4863).
  - D = Type 2024-T6 aluminum alloy bar per SAE AMS-QQ-A-367 or SAE AMS 4133 or Type 2024-T851 aluminum alloy bar per SAE AMS-QQ-A-225/6 or SAE AMS4112.
  - J = Type 304 CRES bar per SAE AMS-QQ-S-763 or SAE AMS5639.
  - K = Type 316 CRES bar per SAE AMS-QQ-S-763 or SAE AMS5648.
  - R = Type 321 CRES bar per SAE AMS-QQ-S-763 or SAE AMS5645.
  - S = 347 CRES forging/bar per SAE AMS-QQ-S-763 or bar per SAE AMS5646.
  - W = Type 7075-T73 aluminum alloy bar per SAE AMS-QQ-A-367 or SAE AMS 4141 or Type 7075-T7351 aluminum alloy bar per SAE AMS-QQ-A-225/9 or SAE AMS4124.
- Table/Type number in this CID (-1 = Table/Type I)
- CID identifier (part of basic part number)

Examples of clean finish PIN's:

AA60051-1-R04P5 = an SAE AS4863 – Type 321 CRES plug, thread size 0.500-14 ANPT, passivated.
AA60051-1-K04P5A = an SAE AS4863 - Type 316 CRES plug, thread size 0.500-14 ANPT, passivated for use in oxygen systems.
AA60051-1-J04P5 = an SAE AS4863 - Type 304 CRES plug, thread size 0.500-14 ANPT, passivated.
AA60051-1-W04J = an SAE AS4863 - Type 7075-T73 aluminum plug, thread size 0.500-14 ANPT, anodized or chemically treated.
AA60051-1-D04JA = an SAE AS4863 - Type 2024-T6 aluminum plug, thread size 0.500-14 ANPT, anodized or chemically treated for use in oxygen systems.

TABLE II. Type II (-2), plug, fitting, port.

A-A-60051 Part Number	Reference Part Number (SAE AS5169)
AA60051-2-( ) 02 (X)	AS5169 ( ) 02 (FC)
AA60051-2-( ) 03 (X)	AS5169 ( ) 03 (FC)
AA60051-2-( ) 04 (X)	AS5169 ( ) 04 (FC)
AA60051-2-( ) 05 (X)	AS5169 ( ) 05 (FC)
AA60051-2-( ) 06 (X)	AS5169 ( ) 06 (FC)
AA60051-2-( ) 08 (X)	AS5169 ( ) 08 (FC)
AA60051-2-( ) 10 (X)	AS5169 ( ) 10 (FC)
AA60051-2-( ) 12 (X)	AS5169 ( ) 12 (FC)
AA60051-2-( ) 16 (X)	AS5169 ( ) 16 (FC)
AA60051-2-( ) 20 (X)	AS5169 ( ) 20 (FC)
AA60051-2-( ) 24 (X)	AS5169 ( ) 24 (FC)
AA60051-2-( ) 28 (X)	AS5169 ( ) 28 (FC)
AA60051-2-( ) 32 (X)	AS5169 ( ) 32 (FC)

(FC) = Finish codes from SAE AS5169, differ from this CID's finish codes unless otherwise indicated.

Material code "T", Titanium alloy, not included in this CID.

Table notes: (continued on next page)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:33Z
Check the source to verify that this is the current version before use.

PIN construction

AA60051 -2- ( ) 16 (X) V

- V — The finish code letter "V" will be placed here only for material code letters "D" and "W" parts that are to be coated with high purity aluminum (see SAE AS5169)
- (X) — Finish code (see 3.3)
- 16 — Size code, tube size in 0.062-inch increments (see SAE AS5169)
- ( ) — Material codes (see SAE AS5169 except as noted):
  - X = Type 4130 alloy steel bar per SAE AMS6370 or Type 4140 per SAE AMS6382 (differs from SAE AS5169).
  - D = Type 2024-T6 aluminum alloy bar per SAE AMS-QQ-A-225/6 or SAE AMS4339 or Type 2024-T851 aluminum alloy bar per SAE AMS-QQ-A-225/6 or SAE AMS4339.
  - J = Type 304 corrosion resistant steel bar per SAE AMS-QQ-S-763 or SAE AMS5639.
  - K = Type 316 corrosion resistant steel bar per SAE AMS-QQ-S-763 or SAE AMS5648.
  - R = Type 321 corrosion resistant steel bar per SAE AMS-QQ-S-763 or SAE AMS5645.
  - S = Type 347 corrosion resistant steel bar per SAE AMS-QQ-S-763 or SAE AMS5646.
  - W = Type 7075-T73 aluminum alloy bar per SAE AMS-QQ-A-225/9 or Type 7075-T7351 aluminum alloy bar per SAE AMS4124.
- -2- — Table/Type number in this CID (-2 = Table/Type II)
- AA60051 — CID identifier (part of basic part number)

Examples of clean finish PIN's:

AA60051-2-X03QX = an SAE AS5169 - carbon steel plug, thread size 0.3750-24 UNJF, Zinc-Nickel Type K16 plating.
AA60051-2-X03QXL = an SAE AS5169 - carbon steel plug, thread size 0.3750-24 UNJF, Zinc-Nickel Type K16 plating with safety wire locking holes.
AA60051-2-J03P5 = an SAE AS5169 - Type 304 CRES, thread size 0.3750-24 UNJF, passivated.
AA60051-2-W03J = an SAE AS5169 - Type 7075-T73 aluminum plug, T thread size 0.500-14, anodized or chemically treated.
AA60051-2-D03LV = an SAE AS5169 - Type 2024-T6 aluminum plug, thread size 0.3750-24 UNJF, coated with high purity aluminum, with safety wire locking holes.

7.2 Source of Government documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/browse/index/far.

7.2.2 Executive order (EO). Copies of EO 14057 are available online from https://www.fedcenter.gov/_kd/go.cfm?destination=ShowItem&Item_ID=38808.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.3 Sources for Non-Government documents.

7.3.1 ASTM documents: Copies of American Society for Testing and Materials (ASTM) International documents are available online from https://www.astm.org/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:33Z
Check the source to verify that this is the current version before use.

7.3.2 SAE documents: Copies of SAE International are available online from https://www.sae.org.

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and date.
b. National stock number, (if available).
c. Material (steel alloy, steel, or aluminum).
d. Part number.
e. Quantity.

Key words:

a. Aluminum
b. CRES
c. Finish
d. Steel
e. Plating
f. Passivate

MILITARY INTERESTS:

Custodian:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AV, MI
Navy – SA
Air Force – 11

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 4730-2023-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:33Z
Check the source to verify that this is the current version before use.